

FINAL EXPLANATION: BIOLOGY GRADE 11

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 2.3 EXCRETION IN PLANTS: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Our driving question is: How do trees and crops clear out their metabolic wastes without any organs or excretory systems, and why are these waste products worth so much money in our markets? Use what you observed on the school nature walks (cut Acacia and Euphorbia, droplets on grass and arrowroot leaves, falling yellow leaves), the guttation experiment, the leaf microscope work, your class tables and your community interview portfolio. Answer every section in your own words in the blank space under each prompt. Use Claim-Evidence-Reasoning (CER) where asked, name the evidence you collected, and write in full sentences. You may add labelled sketches.

Part 1: Why plants do not need kidneys

In Lesson 1 we debated the statement 'Plants do not need kidneys.' (a) Define excretion. (b) Give three reasons why plants can manage without complex excretory organs. (c) Write a short CER statement that answers: why are plants able to live without kidneys, bladders or sweating systems?

(a) Excretion is the removal of metabolic wastes from the body. (b) First, plants have a much lower metabolic rate than animals (shown on the chart), so they make little waste. Second, they recycle gases: the CO₂ from respiration is used for photosynthesis and the O₂ from photosynthesis is used for respiration, as the mesophyll cell diagrams showed, so these do not build up as wastes. Third, they store insoluble wastes such as resin, latex and gum in tissues like bark, vacuoles and old leaves, where they do no harm. (c) Claim: Plants do not need kidneys. Evidence: they have low metabolic rates, they recycle CO₂ and O₂ in their mesophyll cells, and they store products such as gum arabic and latex that we saw oozing from cut Acacia and Euphorbia. Reasoning: because little waste is made and much of it is reused or stored harmlessly, a complex excretory organ is not needed.

Part 2: Routes by which wastes leave the plant

Think about the nature walk, the potted seedling under the bell jar and the plastic bag over a branch. (a) Name the routes or processes by which plants lose wastes that we studied, and say which kind of waste leaves by each. (b) Explain how guttation differs from transpiration, using the droplets on arrowroot and grass leaves on the humid

(a) Diffusion of gases (O₂ and CO₂) through stomata and lenticels; transpiration, which loses water vapour; guttation, which loses liquid droplets with dissolved substances through hydathodes in humid conditions; exudation of gums, resins and latex; and shedding of leaves, bark and fruits (tissue fall), which carries away stored insoluble wastes. (b) Transpiration loses water as invisible vapour, which we only saw as droplets collecting inside the plastic bag over a branch. Guttation loses liquid droplets, carrying dissolved substances, from the edges of leaves through hydathodes in high humidity, as on the arrowroot and grass leaves at dawn. (c) Exudation: the hardened amber gum on Acacia and the milky latex oozing from a cut Euphorbia. Tissue fall: the yellow fallen leaves and the leaf scars left on stems.

morning and what you saw at the plastic bag. (c) Which of our observations on the walk showed exudation, and which showed tissue fall?	
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Part 3: Where insoluble wastes go

In Lesson 3 you collected young green leaves and old yellow or brown leaves and examined them under the light microscope. (a) What wastes can a plant not dissolve and remove in liquid form, and where does it deposit them? (b) Explain, using a CER statement and the observations you recorded in your own microscope work, how shedding old leaves is a form of excretion. (c) How does this idea also help explain the resin or gum on cut Acacia and Euphorbia?	(a) Insoluble wastes such as calcium oxalate crystals, tannins and resins cannot be carried out in solution, so the plant deposits them in non-vital parts such as bark, old leaves, fruits and flowers. (b) Claim: Plants excrete insoluble wastes by depositing them in old or non-vital tissues that are then shed. Evidence: under the microscope we looked for crystals in young green and old yellow leaves and recorded their shape and how they reacted to iodine and dilute acid; old leaves had more crystals than young leaves and the crystals did not dissolve in dilute acid, so the wastes build up in old leaves and are carried away when those leaves fall. Reasoning: locking wastes in tissue the plant no longer needs and then dropping it removes the wastes without any excretory system, so falling leaves are excretion. (c) The gum and resin stored in bark are also wastes kept away from living cells; when the bark is cut they ooze out and harden, which is what we saw on the Acacia and Euphorbia trees around school.
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Part 4: Waste to the plant, wealth to people

Use your class table from Lesson 4 and your community interview portfolio and group poster. (a) Name the types of plant excretory products that we categorised, giving one plant source and one use for each. (b) Why can something that is waste to the plant be valuable in Kenyan markets? Give examples of places or products from our lessons. (c) Some alkaloids are harmful when abused. Describe the risk of such abuse and how people should handle these products.	(a) Latex, from Euphorbia or papaya, used for example as papain from papaya latex and in rubber. Gums and resins, from Acacia, used as gum arabic for sealing, sticking and thickening food. Tannins, from plants such as tea leaves, used in tanning leather. Alkaloids, such as quinine used for malaria and caffeine from tea used in drinks. Essential oils, from lemongrass or eucalyptus, which we smelled when we crushed the leaves. (b) The plant only stores or releases these chemicals as wastes, but people use their chemical properties for medicine, industry and food, so they can be sold. Examples are tea from Kericho, gum arabic from northern Kenya and Turkana collectors, and quinine for malaria. Our interviews also recorded uses, prices and tallies of what community members use. (c) Some alkaloids are habit-forming; abusing them leads to drug and substance abuse, which harms health and can cause addiction. They should only be used in regulated ways, as medicine for the right purpose and amount, and people should be warned of the risks.
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Part 5: Final Explanation

Answer the driving question in your own words as a complete Claim-Evidence-Reasoning explanation: How do trees and crops clear out their metabolic wastes without any organs or excretory systems, and why are these waste products worth so much money in our markets? Refer to the Acacia and Euphorbia trees, the droplets on arrowroot and grass leaves, and the falling old leaves. Then state the new question that our class now wants to investigate and say why it matters.

Claim: Trees and crops clear their wastes without kidneys because they make little waste, reuse much of it, store the rest harmlessly and release it through several surfaces; and these wastes are worth money because people can use their chemical properties. Evidence: plants have low metabolic rates and recycle CO₂ and O₂ in the mesophyll cells. Gases leave by diffusion through stomata and lenticels, and water vapour by transpiration. Liquid wastes leave as guttation droplets from leaf edges of arrowroot and grass on humid mornings. Gums, resins and latex ooze from cut Acacia and Euphorbia. Insoluble wastes such as calcium oxalate crystals, tannins and resins are deposited in non-vital tissues such as bark and old leaves, which are shed, as we saw with the yellow leaves on the ground. Our interviews and tables showed gum arabic, latex, tannins, quinine, caffeine and essential oils being used for sealing, medicine, tanning and drinks, bringing income, for example tea from Kericho. Reasoning: because wastes are small in amount, recycled, stored in non-vital tissue and removed by tissue fall, exudation, guttation, transpiration and diffusion, no kidney-like organ is needed. What is waste to the plant is a useful material to people, so it has market value, although habit-forming alkaloids must be used carefully. New question: How can sustainable harvesting of useful plant excretory products be balanced with environmental conservation? It matters because over-harvesting could damage the trees and the incomes that depend on them.

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining why plants lack excretory organs	Gives all three reasons (low metabolic rate, recycling of gases, storage of insoluble wastes) with specific evidence from the chart and mesophyll diagrams.	Gives two or three reasons with some evidence.	Gives one reason or lists reasons without evidence.
Describing the routes of waste removal	Correctly names diffusion, transpiration, guttation, exudation and tissue fall, matches each to the waste type, and clearly distinguishes guttation from transpiration using local observations.	Names most routes correctly with minor confusion between guttation and transpiration.	Names few routes or confuses them.
Use of evidence and CER reasoning	Claim is clear, evidence comes from lesson observations (Acacia, Euphorbia, arrowroot and grass droplets, leaf microscope work), and reasoning links the evidence to the claim.	Claim and evidence present; reasoning is partial.	Claim is unsupported or evidence is not from the lessons.
Economic value and community uses	Accurately links products, sources and uses (gum arabic, latex, tannins, quinine, caffeine) to Kenyan markets, and explains the risk of harmful alkaloids.	Gives several correct products and uses, with limited explanation of value or risks.	Gives few or inaccurate examples.

Final explanation and new question	Integrates how wastes are removed and why they are valuable in one coherent answer and states the sustainable harvesting question with a reason.	Answer covers both parts but is not fully connected, and the new question is stated.	Answer covers only one part or omits the new question.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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